

AIGR Deep Research Report — *plo1* (S. pombe, UniProt P50528)

Hypothesis slug: `spb-insertion-mechanism` **Seed hypothesis:** *Plo1 directly phosphorylates a spindle-pole-body (SPB) ring component to drive localized nuclear-envelope breakdown and mitotic SPB insertion.* **Focus type:** `free_text` (core-function / mechanistic)

Executive Judgment

Verdict: Partially supported — the process is supported and already curated; the specific "direct phosphorylation of a ring component" molecular step is unresolved (inferred, not demonstrated).

- **Supported (process/BP level):** Plo1 kinase activity is genetically required for the SPB-ring protein redistribution, the localized (partial) nuclear-envelope remodeling, and the insertion of the SPB into the nuclear envelope that build the mitotic spindle. This is exactly the process already captured by GO:0140480 (IMP).
- **Not established (molecular/MF level):** No primary study demonstrates a **direct kinase→substrate** reaction in which Plo1 phosphorylates a *named SPB ring component* to trigger NEBD/insertion. The directness and substrate identity are **inferred from loss-of-activity phenotypes**, not from an in-vitro/in-vivo phospho-site assay.
- **Wording caveat:** S. pombe undergoes **closed mitosis**. "Nuclear-envelope breakdown" should be qualified as **partial/local fenestration** ("insertion into the nuclear envelope"), as the seed's "localized" wording and the GO term already reflect. Global NEBD does **not** occur.
- **Notable constraint:** Plo1 is **not** required for the founding Sad1 ring redistribution (Bestul 2021), so any direct-phosphorylation model cannot be a single upstream Plo1→Sad1 event.

Overall: The hypothesis is a reasonable, literature-consistent *mechanistic model*, but the "directly phosphorylates a SPB ring component" clause exceeds the current evidence. Treat the process claim as **retained/justified**; treat the direct-substrate claim as a **hypothesis requiring a discriminating phospho-assay** before any substrate-specific MF annotation.

Evidence Matrix

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence / limitations
P 34133218 (Bestul et al. 2021, <i>Mol Biol Cell</i>)	Mutant phenotype + super-res localization (SIM)	Supports process; qualifies directness	Plo1 drives localized NEBD & SPB-ring formation for insertion	SPB proteins redistribute into a ring around the SPB (precursor for localized NEBD); Plo1 activity is vital for redistribution of the other SPB ring proteins and for complete NEBD enabling SPB insertion; Plo1 is NOT required for Sad1 redistribution	S. pombe mitosis; plo1 loss-of-activity	Functional necessity, not direct kinase substrate assay ; no phospho-site mapped
P 24963130 (Wälde & King 2014, <i>J Cell Biol</i>)	Mutant phenotype + interaction	Supports process; qualifies directness	Plo1 promotes SPB remodeling/insertion into NE	Plo1 activity drives a burst of Cut12/Pcp1 incorporation; delayed Plo1 recruitment (Kms2 depletion) → SPB-insertion defects; Kms2 binds Cut12/Pcp1/Plo1	S. pombe SUN-KASH (Kms2/Sad1)	Effect partly v recruitment timing; incorporation demonstrated direct phosphorylat
P 7744248 (Ohkura,	Mutant phenotype		plo1 essential	plo1 loss → monopolar spindle /		Pleiotropic (also septation actin ring);

Citation	Evidence type	Direction	Claim tested	Key finding	Context	Confidence / limitations
Hagan & Glover 1995, <i>Genes Dev</i>)		Supports essential role	for bipolar spindle	mitotic arrest; overexpression → monopolar spindles + ectopic septa	<i>S. pombe</i> disruption/overexpression	silent on direct SPB-ring phospho-mechanism
P 23222840 (Grallert et al. 2013, <i>Nature</i>)	Mutant phenotype + localization	Qualifies (context)	Local MPF activates Plo1 at G2 SPB	MPF activation of Plo1 on the SPB dictates timing of mitotic commitment & NETO	<i>S. pombe</i> SPB	Places Plo1 upstream in commitment; not a direct NEBD substrate mechanism
QuickGO record P50528 (76 annotations)	Database	Supports (process already curated)	Existing GO capture	GO:0140480 "mitotic spindle pole body insertion into the nuclear envelope" (IMP); CC: mitotic SPB, half bridge, inner plaque (IDA); MF only generic Ser/Thr kinase activity	GO database	IMP = mutant based; no substrate-level MF annotation

(Provenance CSVs saved during the run: `plo1_evidence_matrix.csv`, `plo1_go_decision_table.csv`.)

GO Curation Implications (leads — require curator verification)

Term	Aspect	Action	Rationale
GO:0140480 mitotic spindle pole body insertion into the nuclear envelope	BP	RETAIN (IMP)	Directly matches the seed's process claim; supported by  34133218 and  24963130 loss-of-activity phenotypes.
GO:0004674 protein serine/threonine kinase activity	MF	RETAIN	Core catalytic activity (EXP/IBA/IDA).
Proposed substrate-specific "phosphorylates SPB ring component" MF/BP	MF	DO NOT ADD (yet)	No direct kinase–substrate assay; adding a specific substrate MF would over-annotate beyond the evidence.
GO:0007052 mitotic spindle organization	BP	RETAIN	Consistent with monopolar-spindle phenotype.

- **Annotation provenance (verified this run):** the plo1 GO:0140480 annotation is a single record — `qualifier=involved_in, evidence=IMP, reference=<svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 16 16" width="14" height="14" class="pubmed-icon" style="display:inline !important;width:14px;height:14px;min-width:14px;min-height:14px;flex-shrink:0;vertical-align:middle;margin-right:3px;"><rect x="1" y="1" width="14" height="14" rx="2" fill="#326599"/><text x="8" y="12" text-anchor="middle" style="font-size:11px;font-weight:bold;font-family:Arial,sans-serif;fill:white;">P</text></svg>34133218, assignedBy=PomBase`. PomBase itself curated this at **mutant-phenotype (IMP) / involved_in** strength — i.e., functional involvement, deliberately stopping short of a direct molecular-activity claim. The seed hypothesis's "directly phosphorylates a specific ring component" therefore **over-specifies** the mechanism relative to both the primary paper and the existing annotation.

- The **BP process** in the hypothesis is already annotated and justified — the hypothesis does **not** warrant a new/stronger BP term.
 - The **MF layer** should stay at generic kinase activity. Do **not** promote to a substrate-specific molecular function, and do **not** fall back to "protein binding" as the informative term.
 - If any curator statement asserts Plo1 *directly phosphorylates* a specific ring component, it should be flagged as **mechanistic inference (IMP-level)**, **not IDA/direct-assay**.
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Mechanistic Scope

- **Immediate molecular activity (direct):** ATP-dependent protein serine/threonine kinase (Polo-like kinase) — well established.
 - **Directly demonstrated at the SPB:** Plo1 **localizes** to the SPB (IDA) and its **catalytic activity is required** for mitotic SPB remodeling/insertion (loss-of-activity phenotype).
 - **Inferred (not directly shown):** that a *specific SPB ring protein* is a **direct Plo1 phosphosubstrate**, and that this phospho-event is the trigger for local fenestration/insertion.
 - **Downstream/pleiotropic (do not conflate with the core insertion mechanism):** mitotic commitment (Cdc25/Wee1 axis), septation-initiation-network activation, cytokinetic ring, NETO/polarity, meiotic SPB remodeling. Plo1's essentiality phenotypes are broad, so an SPB-insertion defect in a plo1 mutant is not, by itself, proof of a direct ring-substrate event.
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Conflicts and Alternatives

1. **Directness gap / alternative = indirect action:** Plo1 could enable insertion by controlling **recruitment/incorporation timing** of SPB components (Wälde & King: "burst of incorporation") rather than by directly phosphorylating a ring protein — the phenotype is consistent with either.
2. **Sad1 independence:** Plo1 is dispensable for Sad1 ring redistribution (Bestul 2021), so the "ring component" in the hypothesis cannot be Sad1; the responsible substrate (if any) is unidentified.
3. **Wording/organism biology:** "nuclear-envelope breakdown" risks importing an open-mitosis concept; *S. pombe* uses **partial/local fenestration** in an otherwise closed mitosis

(cf. Brr6 fenestra work,

P 22042620

The term should stay "insertion into the nuclear envelope."

4. **Pleiotropy / phenotype attribution:** plo1 null is pleiotropic (spindle + septation), so SPB-insertion phenotypes must be isolated from indirect commitment/SIN effects.
5. **No paralog-overannotation issue detected:** Plo1 is the single *S. pombe* Polo kinase; the concern here is *directness/substrate identity*, not paralog confusion.

Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Identity of the direct Plo1 substrate among SPB ring proteins	Searched primary literature + QuickGO MF annotations	Determines whether a substrate-specific MF/mechanism can be curated	In-vitro kinase assay + in-vivo phospho-site mapping (phosphoproteomics) on candidate ring proteins
Whether phosphorylation <i>causes</i> insertion vs. correlates	Bestul/Wälde phenotypes	Distinguishes direct trigger from permissive/recruitment role	Non-phosphorylatable (S/T→A) and phosphomimetic (S/T→D/E) substrate alleles rescuing/blocking insertion
Kinase-dead vs. localization-only separation	Grallert 2013 (activity timing)	Separates catalytic requirement from scaffolding	Analog-sensitive plo1 (as-plo1) acute inhibition during SPB insertion window
Exact NE-remodeling target (lipid/pore/fenestra machinery)	Brr6/Tts1 literature noted	Connects Plo1 activity to membrane fenestration	Epistasis of plo1 with brr6/apq12/tts1 for insertion

Discriminating Tests

1. **Analog-sensitive plo1 (as-plo1) acute inhibition** timed to the SPB-insertion window + live imaging of SPB/NE markers — tests catalytic necessity for insertion specifically (vs. earlier commitment).
 2. **Candidate-substrate phospho-mapping:** IP-MS phosphoproteomics of SPB ring proteins (e.g., Cut12, Pcp1, Cut11, other ring proteins from Bestul 2021) ± active Plo1; confirm sites by in-vitro kinase assay with recombinant Plo1.
 3. **Phosphosite allele swap:** non-phosphorylatable vs. phosphomimetic alleles of the top candidate; score SPB insertion/fenestration and bipolar spindle formation.
 4. **Epistasis** of plo1 activity with brr6/apq12/tts1 fenestration mutants to place Plo1 relative to NE-membrane remodeling.
 5. **Comparative check** with budding-yeast Polo (Cdc5)/Mps1 SPB-insertion pathways to test conservation of a direct ring-substrate mechanism.
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Curation Leads (require curator verification)

- **Action:** Retain BP GO:0140480 (IMP) and MF GO:0004674; **do not** add a substrate-specific "phosphorylates SPB ring component" MF/BP term at this time; **do not** downgrade to "protein binding."
- **Candidate references + snippets to verify:**
- PMID 34133218: *"its activity is vital for redistribution of other SPB ring proteins and for complete NEBD at the SPB to allow for SPB insertion"* and *"the Polo kinase Plo1 is not necessary for Sad1 redistribution"*.
- PMID 24963130: *"Plo1 activity supports mitotic SPB remodeling by driving a burst of incorporation of Cut12 and Pcp1"*.
- PMID 7744248 (essentiality / monopolar spindle); PMID 23222840 (SPB-local Plo1 activation timing).
- **Suggested curator questions:**
- Is any statement in the review implying **direct phosphorylation** of a specific SPB ring component? If so, mark as **mechanistic inference**, not direct assay.

- Should the review text replace "nuclear-envelope breakdown" with "**localized/partial nuclear-envelope fenestration / SPB insertion into the nuclear envelope**" to match closed-mitosis biology?
- **Suggested experiments:** the discriminating tests above (as-plo1 timing; phospho-site mapping + S/T→A/D alleles).

Limitations of this analysis

- Based on abstract-level literature synthesis and the public GO/QuickGO record; full-text methods were not parsed.
- No local sequence/structure computation was needed to adjudicate the core (directness) question; the decisive gap is an unavailable **direct kinase-substrate assay**, which cannot be substituted by bioinformatics.
- One citation-snippet auto-validation failed only because the stored abstract copy for

P 34133218

was truncated; the verbatim text above is from the full NCBI abstract retrieved during the run.